Type MS · Compression: lossy or lossless?

Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

5(a)	Award **one** mark for each correct column: Translator Characteristic / Assembler / Interpreter / Compiler Does **not** have to be present in memory to execute the translated code. — ✓ — — ✓ Translates high level language source code. — — ✓ ✓ Executes high level code until either the program has finished, or a syntax error is encountered. — — ✓ — Translates low level language source code into machine code. — ✓ — —		3
5(b)	Award **two** marks for a linked explanation such as: • Fewer instructions are executed per second (1) so gameplay will not appear to execute as smoothly / graphics lag / fewer frames per second / can only handle less complex graphics (1) • Consoles will execute more instructions per second / have faster processors / GPUs (1) so gameplay will not be realistic / mirrored / fair comparison (1)	One mark can be awarded processing speed implication, but the second mark is dependent on having a gameplay impact being identified. e.g. "A lower clock speed means processing will take longer" is generic for one mark. Low performance on its own is not enough.	2

16

Type MS · Compression: lossy or lossless?

Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

<table>
<tr>
<td>5(c)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Issue</th><th>Utility software</th></tr>
<tr><td>Her word processor cannot read all of the files because it is corrupt.</td><td>File repair/recovery</td></tr>
<tr><td>She has a JPEG image and needs to send it to a publisher who can only read PNG image files.</td><td>Converting</td></tr>
<tr><td>There has been a flood and files on a server have been damaged, but Anna needs to access her data.</td><td>Backup/restore</td></tr>
<tr><td>Anna's hard drive is running very slowly when she tries to access a file.</td><td>Defragmentation/defrag</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>5(d)</td>
<td>Award one mark for:
• Lossless (1)

Award one mark for any of the following up to a maximum of two marks:
• No data will be lost / the process is reversable (1)
• The manual will still be readable (1)
• Removal of data would change the meaning of text (1)
• Proofreader will be able to edit the text (1)</td>
<td>Allow converse if candidate gives reasons why lossy would not be used for justification

Allow RLE as a valid lossless compression method.</td>
<td>3</td>
</tr>
</table>

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Type MS · Compression: lossy or lossless?

Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

5(e)	Award **one** mark for each of: Method 1 • (C AND NOT D) (1) • (NOT C AND D) (1) • <expr1> OR <expr2> (1) Method 2 • (C OR D) (1) • NOT (C AND D) (1) • <expr1> AND <expr2> (1) Example statements: (C AND NOT D) OR (NOT C AND D) (C OR D) AND NOT (C AND D)	Order of precedence is NOT, AND, OR Note other equivalent solutions MAX 2 if all terms correct apart from missing brackets that are required. Do not credit circuit diagrams.	3
		Total for question 5	**15**

Type MS · Compression: lossy or lossless?

Answer 1 of 9 · Question 5 part d · 3 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Compression: lossy or lossless?

Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part c.i in this mark scheme extract

4(a)	Any **two** from: Kilobyte is a SI (physics/science) / International System of Units measurement, whereas Kibibyte is an IEC (computing) / International Electrotechnical Commission / International System of Quantities / ISO measurement (1) Kilobyte is equivalent to 1000 bytes, whereas a kibibyte is equivalent to 1024 bytes (1) Kilobyte is equivalent to 10^3 bytes, whereas a kibibyte is equivalent to 2^{10} bytes (1) Kilobyte is a base 10 measurement, whereas a kibibyte is a base 2 measurement (1)	Ignore units	2

Type MS · Compression: lossy or lossless?

Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **4 marks** for 19.17 or **One** mark for each of: • Calculates correct bits/bytes(1) • Adds metadata (1) • Calculates correct megabytes (1) • Completely correct response (1) Examples: $\frac{\frac{2322 \times 4128 \times 16}{8} + 975}{1000 \times 1000}$ $\frac{(2322 \times 4128 \times 2) + 975}{1000 \times 1000}$ ((2322 x 4128 x 16 bits per colour) ÷ 8 bits per byte) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB ((2322 x 4128 x 2 bytes per colour) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB	• Arithmetic precedence rules must be clearly followed • Units are not required • Equivalent expressions are accepted • Calculations not explicit (8x1000x1000) but expressed (8000000) gain the mark	4

Type MS · Compression: lossy or lossless?

Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Any **two** from: • To save disc/storage space (1) • Reduce transmission time (1) • Meet email attachment size restrictions (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Any **two** from: • Compressing / decompressing the file takes execution time (1) • Both compressing and decompressing have to be done with compatible software / both ends of transmission have to use compatible software (1) • (If the compression utility uses a lossy algorithm then) some of the data will be permanently removed / it won't be possible to reconstruct the original file. (1) • Can affect the quality of the file (1) • File size can increase (1)		2

Type MS · Compression: lossy or lossless?

Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award 4 marks for 11 Award 3 marks for a value between 10 and 11 **One** mark for each: • 80 x 66 x 2 or 80 x 66 x 16 ÷ 8 (1) • + 40 or + 80 (1) • ÷ 1024 (1) • Ceiling or RoundUp (1)	• Arithmetic precedence rules must be clearly followed • Units are not required • Alternative notations for ceiling are accepted • If 16 bits per Unicode character is not awarded, then allow follow through error • Allow follow through error, if interim calculations attempted • Equivalent expressions are accepted	4

Type MS · Compression: lossy or lossless?

Answer 2 of 9 · Question 4 part c.i · 2 marks

2019 JUN · Paper 01 · extract 5 of 5 · find part c.i in this mark scheme extract

Question Number	Answer	Mark

Type MS · Compression: lossy or lossless?

Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part c.ii in this mark scheme extract

4(a)	Any **two** from: Kilobyte is a SI (physics/science) / International System of Units measurement, whereas Kibibyte is an IEC (computing) / International Electrotechnical Commission / International System of Quantities / ISO measurement (1) Kilobyte is equivalent to 1000 bytes, whereas a kibibyte is equivalent to 1024 bytes (1) Kilobyte is equivalent to 10^3 bytes, whereas a kibibyte is equivalent to 2^{10} bytes (1) Kilobyte is a base 10 measurement, whereas a kibibyte is a base 2 measurement (1)	Ignore units	2

Type MS · Compression: lossy or lossless?

Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **4 marks** for 19.17 or **One** mark for each of: • Calculates correct bits/bytes(1) • Adds metadata (1) • Calculates correct megabytes (1) • Completely correct response (1) Examples: $\frac{\frac{2322 \times 4128 \times 16}{8} + 975}{1000 \times 1000}$ $\frac{(2322 \times 4128 \times 2) + 975}{1000 \times 1000}$ ((2322 x 4128 x 16 bits per colour) ÷ 8 bits per byte) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB ((2322 x 4128 x 2 bytes per colour) + 975 bytes --- 1000 bytes per KB x 1000 kilobytes per MB	• Arithmetic precedence rules must be clearly followed • Units are not required • Equivalent expressions are accepted • Calculations not explicit (8x1000x1000) but expressed (8000000) gain the mark	4

Type MS · Compression: lossy or lossless?

Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Any **two** from: • To save disc/storage space (1) • Reduce transmission time (1) • Meet email attachment size restrictions (1)		2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Any **two** from: • Compressing / decompressing the file takes execution time (1) • Both compressing and decompressing have to be done with compatible software / both ends of transmission have to use compatible software (1) • (If the compression utility uses a lossy algorithm then) some of the data will be permanently removed / it won't be possible to reconstruct the original file. (1) • Can affect the quality of the file (1) • File size can increase (1)		2

Type MS · Compression: lossy or lossless?

Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(d)	Award 4 marks for 11 Award 3 marks for a value between 10 and 11 **One** mark for each: • 80 x 66 x 2 or 80 x 66 x 16 ÷ 8 (1) • + 40 or + 80 (1) • ÷ 1024 (1) • Ceiling or RoundUp (1)	• Arithmetic precedence rules must be clearly followed • Units are not required • Alternative notations for ceiling are accepted • If 16 bits per Unicode character is not awarded, then allow follow through error • Allow follow through error, if interim calculations attempted • Equivalent expressions are accepted	4

Type MS · Compression: lossy or lossless?

Answer 3 of 9 · Question 4 part c.ii · 2 marks

2019 JUN · Paper 01 · extract 5 of 5 · find part c.ii in this mark scheme extract

Question Number	Answer	Mark

Type MS · Compression: lossy or lossless?

Answer 4 of 9 · Question 4 part c · 1 mark

2020 NOV · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

4(a)(i)	Digital (signal) / converted analogue (signal) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(ii)	Analogue (signal) / analogue (sound wave) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(iii)	2 Hertz / hertz /Hz (1)	• Do not penalise spelling • Accept 2 samples/cycles per second	1

Question Number	Answer	Additional Guidance	Mark
4(a)(iv)	Any **one** mark for: • The (analogue sound) wave will be represented more accurately / the fidelity/quality of the recording is improved (1) • The file size will increase / more data stored (as each sample takes up disk space) (1)		1

Question Number	Answer	Additional Guidance	Mark
4(a)(v)	• 3 (1) • The range of sampled sound is from 0-6. To store 7 distinct values, you need 3 bits (111 binary = 7 denary) (1) OR graph shows 8 possible values, 3 bits can store 0 - 7 / 8 values (1)		2

Question Number	Answer	Additional Guidance	Mark

Type MS · Compression: lossy or lossless?

Answer 4 of 9 · Question 4 part c · 1 mark

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4(a)(vi)	Correct binary value 101 (1)	Ignore leading zeros	1

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Type MS · Compression: lossy or lossless?

Answer 4 of 9 · Question 4 part c · 1 mark

2020 NOV · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
4(b)	819 seconds * 8 bytes * 47,000 hertz 819 x 8 x 47000 819 seconds * 64÷8 bytes * 47,000 hertz 818 x 64 x 47000 8 819 (1) 8 or 64÷8 (1) 47,000 (1) Fully correct expression (1)	Award equivalent expressions	4

Question Number	Answer	Additional Guidance	Mark
4(c)	Any **one** from: • The sound quality may be poorer/lower (for some people) (1) • Cannot get the original back after compression (1)		1

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Type MS · Compression: lossy or lossless?

Answer 4 of 9 · Question 4 part c · 1 mark

2020 NOV · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Compression: lossy or lossless?

Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 5 · find part f.i in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · Compression: lossy or lossless?

Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 5 · find part f.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · Compression: lossy or lossless?

Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 3 of 5 · find part f.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · Compression: lossy or lossless?

Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 4 of 5 · find part f.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · Compression: lossy or lossless?

Answer 5 of 9 · Question 2 part f.i · 2 marks

2021 JUN · Paper 01 · extract 5 of 5 · find part f.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Compression: lossy or lossless?

Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part f.ii in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · Compression: lossy or lossless?

Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part f.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · Compression: lossy or lossless?

Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part f.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · Compression: lossy or lossless?

Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part f.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · Compression: lossy or lossless?

Answer 6 of 9 · Question 2 part f.ii · 1 mark

2021 JUN · Paper 01 · extract 5 of 5 · find part f.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Compression: lossy or lossless?

Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part b.iii in this mark scheme extract

1(ai)	0100 1101 100 1101 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(ii)	1000 0010 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(iii)	Award **one** mark for each of: • C (1) • 6 (1)	Must be in the correct order	2

Type MS · Compression: lossy or lossless?

Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	400 pixels x 200 pixels x 12 bits / 8 bits per byte Allow expression in word form $\frac{400 \times 200 \times 12}{8}$ 400 x 200 **OR** 80000(1) 12/8 **OR** 1.5 (1)	Accept any other use of 400, 200, 12, 8 that gives the correct answer Allow one mark for 120,000 with no expression	2
1(b)(ii)	Spaces are to help legibility. Award **three** marks for: Line 1 3W 1B 4W Line 2 1W 6B 1W Line 3 1B 2W 1B 1W 1B 2W Line 4 3W 2B 3W Award **two** marks for any three lines correctly encoded Award **one** mark for any two lines correctly encoded	Allow **one** error in the sequence Accept letter before number but must be consistently used	3
1(b)(iii)	Award **one** mark from: • None of the original (image) detail/quality is lost (when the image is stored) (1) • None of the data/information is (permanently) removed (1) • It can be decompressed without losing detail (1)		1

Type MS · Compression: lossy or lossless?

Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award up to **three** marks for a linked description: • set the sample rate/parameters/bit-depth (1) • sample (the analogue sound) (1) • measure the sound amplitude/volume/frequency (1) • give a (binary) value/number for each measurement (1) • store data as sample rate and values / digital signals (1)		3
1(c)(ii)	Award **one** mark from: • sound may be missing some frequencies (1) • some audio information/data is lost/deleted during the (compression) process (1)		1
		Total	**16**

Type MS · Compression: lossy or lossless?

Answer 7 of 9 · Question 1 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Compression: lossy or lossless?

Answer 8 of 9 · Question 1 part c.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

1(ai)	0100 1101 100 1101 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(ii)	1000 0010 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(iii)	Award **one** mark for each of: • C (1) • 6 (1)	Must be in the correct order	2

Type MS · Compression: lossy or lossless?

Answer 8 of 9 · Question 1 part c.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	400 pixels x 200 pixels x 12 bits / 8 bits per byte Allow expression in word form $\frac{400 \times 200 \times 12}{8}$ 400 x 200 **OR** 80000(1) 12/8 **OR** 1.5 (1)	Accept any other use of 400, 200, 12, 8 that gives the correct answer Allow one mark for 120,000 with no expression	2
1(b)(ii)	Spaces are to help legibility. Award **three** marks for: Line 1 3W 1B 4W Line 2 1W 6B 1W Line 3 1B 2W 1B 1W 1B 2W Line 4 3W 2B 3W Award **two** marks for any three lines correctly encoded Award **one** mark for any two lines correctly encoded	Allow **one** error in the sequence Accept letter before number but must be consistently used	3
1(b)(iii)	Award **one** mark from: • None of the original (image) detail/quality is lost (when the image is stored) (1) • None of the data/information is (permanently) removed (1) • It can be decompressed without losing detail (1)		1

Type MS · Compression: lossy or lossless?

Answer 8 of 9 · Question 1 part c.ii · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award up to **three** marks for a linked description: • set the sample rate/parameters/bit-depth (1) • sample (the analogue sound) (1) • measure the sound amplitude/volume/frequency (1) • give a (binary) value/number for each measurement (1) • store data as sample rate and values / digital signals (1)		3
1(c)(ii)	Award **one** mark from: • sound may be missing some frequencies (1) • some audio information/data is lost/deleted during the (compression) process (1)		1
		Total	**16**

Type MS · Compression: lossy or lossless?

Answer 8 of 9 · Question 1 part c.ii · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

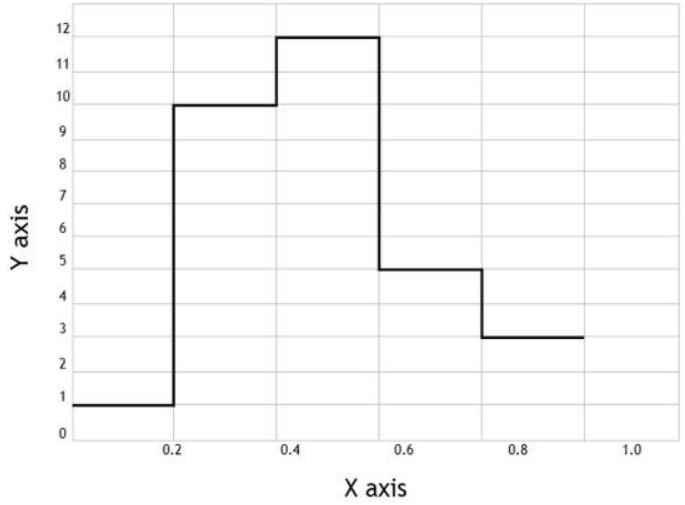
Type MS · Compression: lossy or lossless?

Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:

• At least four conversions plotted at the correct amplitude (1)
• Correct start point (1) 0 – 1 digital. 0 – 0 analogue
• Digital sound wave drawn (1) using candidate's plots

<table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table>
 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:

• Time (1)
• Sample interval/period (1)
• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:

• Amplitude / sound level / volume (1)
• Metres/centimetres/nanometres (1)
• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

Type MS · Compression: lossy or lossless?

Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:
Transfer time would be longer (1)
Larger file size / less compression (1)
Takes up more storage space (1)
Uses more of her data allocation (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:
Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)
Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)
Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)
Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:
Storage host could be targeted by hackers (1)
Alyssa has less control over her files (1)
An untrustworthy employee (of the cloud storage provider) could steal her files (1)
Reliant on the storage provider for security / safekeeping (1)
Files could be intercepted/corrupted during upload/download (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · Compression: lossy or lossless?

Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

	ru2.mp3	File/media/resource wanted (on the website)/ path / part of path (1)		
			Total for question 3	**13**

Type MS · Compression: lossy or lossless?

Answer 9 of 9 · Question 3 part b.i · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question Number	Answer (flow chart replaced)	Additional Guidance	Mark